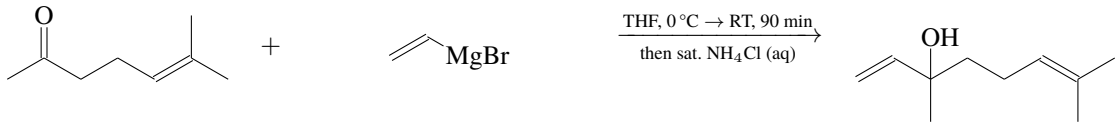
ORGANIC EXPERIMENT 4, PART 1 — PRE-LAB CHECKLIST

Synthesis of (±)-Linalool by Grignard Addition on a Schlenk Line

Condensed from the CM3191 Organic Lab Manual AY26/27 (Dr Hoang T. G., pp. 19–24, 27) and lecture CM3191 Org L4.

Week 9, **Thursday 15 October 2026**, Thursday lab (Dr Hoang T. G.), **Group A** (Schlenk line this week; reduction in Week 10). Venue **[CONFIRM: not stated in the organic schedule]**.



6-methyl-5-hepten-2-one
0.44 mL, 2.98 mmol

vinylMgBr
1 M in THF, 7.5 mL (2.5 equiv)

(±)-linalool
0.460 g theor.

1. Reagent table (manual values; blanks in the manual filled by calculation)

Reagent	MW / conc.	Density	Amount	mmol	Equiv.	Role
6-Methyl-5-hepten-2-one	126.20	0.855 g/mL	0.44 mL (0.376 g)	2.98	1.00	ketone (limiting)
Vinylmagnesium bromide	131.25; 1.0 M in THF	—	7.5 mL	7.5	2.52	nucleophile
THF (anhydrous)	72.11	—	5 mL	—	—	solvent
Sat. NH_4Cl (aq)	—	—	10 mL	—	—	quench
(±)-Linalool	154.25	—	**0.460 g (theor.)**	2.98	—	product

The manual's table rounds the ketone to 3.0 mmol; from 0.44 mL × 0.855 g/mL it is 2.98 mmol. Reaction volume ≈ 12.9 mL in a 50 mL flask.

Recalculate with your actual volume: $m_{\text{theor}} = V_{\text{ketone}} \times 0.855 \times (154.25/126.20) = \mathbf{1.045}\,\boldsymbol{V}_{\text{ketone}}$ (g, with V in mL).

2. Bring / prepare before the session

☐ Official datasheet (p. 6 here = manual p. 27), **printed**
☐ Lab coat, safety goggles, nitrile gloves, covered shoes
☐ Ruler + pencil: TLC plate must be **drawn with distances (cm)**
☐ Re-read L4: stopcock positions, **slow** backfill (bubbler suck-back)
☐ Schlenk-line port **[CONFIRM: assigned port; whether shared with D1]**
☐ This bench sheet (pp. 2–7)
☐ Phone for photos of every volume and balance reading
☐ Marker and labels for the crude flask (kept to Week 11)
☐ Know: **no flammables in the hood** while the heat gun is on
☐ Crude storage until Week 11 **[CONFIRM: where and how]**

3. Session timeline (estimate)

#	Stage	Approx. time
1	Connect two-neck flask to the line; heat-gun drying under vacuum, cool, backfill N_2 (×2)	20 min
2	Add ketone + anhydrous THF; cool in ice-water bath	15 min
3	Purge syringe, withdraw 7.5 mL vinylMgBr under N_2, add dropwise	15 min
4	Stir at RT — *meanwhile: quench syringe, pre-weigh 250 mL rbf, set up TLC chamber and separatory funnel*	**90 min (fixed)**
5	Quench with sat. NH_4Cl (ice bath); extract EtOAc ×3; brine wash	30 min
6	Dry over Na_2SO_4 (15 min) and run TLC ($KMnO_4$) at the same time	20 min
7	Filter, rotary-evaporate, weigh crude; store for Week 11; datasheet photo check and GTA signature	25 min

SAFETY — from the manual (pp. 20–23) and L4

6-Methyl-5-hepten-2-one is a flammable liquid and an eye irritant. Vinylmagnesium bromide is toxic, causes skin irritation and serious eye damage, and **releases flammable gas with water**. Linalool is a skin and eye irritant. Silica gel is toxic if inhaled (Week 11). Ethyl acetate, THF and diethyl ether are highly flammable; hexane is flammable and a neurotoxin. The **hot air gun reaches 300–500 °C**: no flammable chemicals or solvents in the fume hood while heating. Be careful with needles; used syringes go in the **sharps bin** after the residual Grignard is quenched (sat. NH_4Cl, waste fume hood; ethylene forms). L4: backfill an evacuated flask **slowly** so bubbler oil is not sucked into the line; never pull air through a liquid-N_2 cold trap (liquid O_2 risk).

Added notes (not in the manual). The excess Grignard (≈4.5 mmol) releases roughly **110 mL of ethylene** when quenched, and the quench is exothermic: do not quench in a sealed flask — remove or vent the septum first and add NH_4Cl dropwise in the ice bath. Vent the separatory funnel often (EtOAc). Waste: aqueous layers (Mg salts) → aqueous waste; EtOAc / THF → non-halogenated organic waste; needles → sharps. **[CONFIRM: waste bins in the organic lab]**

BENCH MANUAL (ENGLISH)

Grignard Reaction, Part 1: Reaction, Work-up and TLC

Tick each box as you complete the step. "Why" notes explain the purpose; they are not extra steps. Column chromatography is Part 2 (Week 11).

1.1 Drying the apparatus on the Schlenk line (Appendix A, panel 1)

☐ 1. Clear the fume hood of **all flammable chemicals and solvents**. *The heat gun reaches 300–500 °C.*

☐ 2. Put a stir bar in a **50 mL two-necked rbf**. Fit the **side neck with a rubber septum**; connect the **vertical neck to a stopcock adapter** on the Schlenk line. Check that N_2 is bubbling through the bubbler.

☐ 3. Turn the two-way valve to **vacuum**. Heat the flask all over with the **hot air gun** to drive out moisture.

Why: a thin film of water on glass destroys the Grignard reagent (RMgBr + H_2O → RH + Mg(OH)Br) and lowers the yield.

☐ 4. Let the flask cool for a few minutes under vacuum, then turn the valve to **N_2 — slowly**, watching the bubbler.

Why: opening an evacuated flask too fast pulls bubbler oil back into the manifold (L4).

☐ 5. **Repeat once more**: vacuum + heat gun → cool → N_2. **[CONFIRM: the manual gives 2 cycles; L4 recommends 3 vacuum–refill cycles]**

☐ 6. Let the flask cool to **room temperature** before adding any chemicals.

1.2 Reaction

☐ 7. Add **0.44 mL 6-methyl-5-hepten-2-one** (3 mmol) and **5 mL anhydrous THF** to the flask. Photograph the volumes. Cool in an **ice-water bath for 5–10 min**.

☐ 8. Clamp the **vinylMgBr bottle** (1 M in THF) on a retort stand and put it **under N_2** (balloon or Schlenk line).

☐ 9. **Purge a 20 mL syringe with long needle** with N_2 (L4: three times).

☐ 10. Withdraw **7.5 mL** of the Grignard solution. **Before pulling the needle out of the bottle, draw in a little N_2.**

Why: the N_2 plug protects the reagent at the needle tip from air and stops it dripping during the transfer.

☐ 11. Add the vinylMgBr **dropwise** through the septum into the cooled, stirred flask. Record start and end times.

Why: the addition is exothermic; slow addition at 0 °C favours attack on C=O over deprotonation of the α-H (the enolate returns ketone on work-up).

☐ 12. **Remove the ice bath**; stir at room temperature for **90 min**. Start the timer and record the time.

☐ 13. While it stirs: **quench the residual Grignard** in the syringe and needle with sat. NH_4Cl **in the waste fume hood** (ethylene forms); syringe → **sharps bin**. Pre-weigh a dry **250 mL rbf**; set up the TLC chamber (hexane:EtOAc **9:1**) and separatory funnel.

2. Work-up

☐ 14. Turn the two-way valve to **OFF**; disconnect the flask; close it with rubber septa; take it back to **your own fume hood**.

☐ 15. Cool the flask in an **ice-water bath**. Add **10 mL sat. NH_4Cl dropwise**. *Vent first (not in the manual): about 110 mL of ethylene forms.*

Why: NH_4Cl is a weak acid: it protonates the magnesium alkoxide to give linalool and destroys excess vinylMgBr. A strong acid is avoided because linalool is a tertiary allylic alcohol that easily loses water or rearranges under acid.

☐ 16. Add **10 mL water**; transfer the mixture to a **150 mL separatory funnel**. Rinse the flask with **20 mL EtOAc** and add the rinse to the funnel.

☐ 17. Shake (vent often) and separate. Collect the **aqueous** and **organic** layers in separate conical flasks. *EtOAc (0.90 g/mL) is the top layer.*

☐ 18. Return the aqueous layer to the funnel; extract **2 more times with 20 mL EtOAc** (3 extractions in total). Combine the EtOAc layers (≈60 mL).

☐ 19. Return the combined organic layer to the funnel; wash with **15 mL brine**. Put the organic layer in a **dry 250 mL conical flask**.

Why: brine pulls dissolved water out of the EtOAc, so the drying agent has less to do.

☐ 20. Add **anhydrous Na_2SO_4**; let stand **15 min**.

☐ 21. **TLC** (while drying): spot the **starting ketone** and the **reaction mixture** (from the drying solution). Elute with **hexane:EtOAc 9:1**; stain with **$KMnO_4$**. Draw the plate with all distances (cm).

Why: neither compound is UV-active, but both have a C=C that $KMnO_4$ oxidises (yellow-brown spot on purple). The alcohol should run lower than the ketone. This plate sets the eluent for the Week-11 column.

☐ 22. Filter into the **pre-weighed 250 mL rbf**; remove the solvent on the **rotary evaporator** → crude linalool.

☐ 23. Weigh **rbf + crude**; photograph; fill in the **crude product** section of the datasheet (p. 27).

☐ 24. Label and store the crude for Week 11. **[CONFIRM: storage location; stoppered or under N_2]**

☐ 25. Show photos to the GTA; get the **printed official datasheet** checked/signed. **[CONFIRM: sign now, or after the purified mass in Week 11]**

실험 매뉴얼 (한국어)

Grignard 반응 1부: 반응, 후처리, TLC

단계를 마칠 때마다 체크하세요. "이유"는 목적 설명이며 추가 단계가 아닙니다. 컬럼 크로마토그래피는 2부(11주차)입니다.

1.1 Schlenk line에서 장치 건조 (부록 A, 그림 1)

☐ 1. 후드 안의 **인화성 시약과 용매를 모두 치웁니다**. 열풍기*(hot air gun)*는 *300–500 °C*까지 올라갑니다.

☐ 2. **50 mL 2구 둥근바닥 플라스크**에 교반 막대를 넣습니다. **옆 입구는 고무 셉텀**으로 막고, **수직 입구는 Schlenk line의 stopcock adapter**에 연결합니다. 버블러로 N_2가 나오고 있는지 확인합니다.

☐ 3. 2방향 밸브를 **진공**으로 돌리고, **열풍기**로 플라스크 전체를 가열해 수분을 날립니다.
이유*:* 유리 표면의 얇은 물막이 Grignard 시약을 분해합니다(RMgBr + H_2O → RH + Mg(OH)Br). 그만큼 수율이 떨어집니다.

☐ 4. 진공 상태로 몇 분 식힌 뒤, 버블러를 보면서 밸브를 **N_2로 천천히** 돌립니다.
이유*:* 진공인 플라스크를 너무 빨리 열면 버블러 오일이 매니폴드로 빨려 들어갑니다(L4).

☐ 5. **한 번 더 반복**합니다: 진공 + 열풍기 → 식힘 → N_2. **[CONFIRM: 매뉴얼은 2회, L4 강의는 3회 권장]**

☐ 6. 시약을 넣기 전에 플라스크를 **실온까지** 식힙니다.

1.2 반응

☐ 7. 플라스크에 **6-methyl-5-hepten-2-one 0.44 mL**(3 mmol)와 **무수 THF 5 mL**를 넣습니다. 부피는 사진으로 남깁니다. **얼음물 수조에서 5–10분** 식힙니다.

☐ 8. **vinylMgBr 병**(THF 중 1 M)을 스탠드에 고정하고 **N_2 아래**에 둡니다(풍선 또는 Schlenk line).

☐ 9. 긴 바늘을 끼운 **20 mL 주사기를 N_2로 퍼지**합니다(L4: 3회).

☐ 10. Grignard 용액 **7.5 mL**를 뽑습니다. **바늘을 병에서 빼기 전에 N_2를 조금 더 빨아들입니다.**
이유*:* N_2 마개가 바늘 끝의 시약을 공기로부터 보호하고, 옮기는 동안 방울이 떨어지지 않게 합니다.

☐ 11. vinylMgBr를 셉텀을 통해 식혀 둔 교반 중인 플라스크에 **한 방울씩** 넣습니다. 시작과 끝 시각을 기록합니다.
이유*:* 첨가는 발열 반응입니다. 0 °C에서 천천히 넣어야 α-H를 떼는 반응(에놀화, 후처리 후 케톤으로 돌아감)보다 C=O 첨가가 우세합니다.

☐ 12. **얼음 수조를 치우고** 실온에서 **90분** 교반합니다. 타이머를 켜고 시각을 기록합니다.

☐ 13. 교반하는 동안: 주사기와 바늘에 남은 Grignard를 **폐액 후드에서** 포화 NH_4Cl로 **quench**합니다(에틸렌 발생). 주사기는 **sharps 통**에 버립니다. 마른 **250 mL 둥근바닥 플라스크 무게**를 재 두고, TLC 전개조(헥세인:EtOAc **9:1**)와 분별 깔때기를 준비합니다.

2. 후처리

☐ 14. 2방향 밸브를 **OFF**로 돌리고 플라스크를 분리합니다. 고무 셉텀으로 막아 **본인 후드**로 가져옵니다.

☐ 15. 플라스크를 **얼음물 수조**에 넣고 **포화 NH_4Cl 10 mL를 한 방울씩** 넣습니다. 먼저 셉텀을 열거나 벤트하세요 *(매뉴얼에 없음):* 에틸렌이 약 *110 mL* 나옵니다.
이유*:* NH_4Cl은 약산이라 마그네슘 알콕사이드를 양성자화해 linalool을 만들고, 남은 vinylMgBr를 분해합니다. 강산을 쓰지 않는 이유는 linalool이 3차 알릴 알코올이어서 산에서 쉽게 탈수되거나 재배열되기 때문입니다.

☐ 16. **물 10 mL**를 넣고 **150 mL 분별 깔때기**로 옮깁니다. 플라스크를 **EtOAc 20 mL**로 헹궈 깔때기에 더합니다.

☐ 17. 흔들고(자주 벤트) 층을 나눕니다. **수층**과 **유기층**을 각각 다른 삼각플라스크에 받습니다. *EtOAc(0.90 g/mL)*가 위층입니다.

☐ 18. 수층을 다시 깔때기에 넣고 **EtOAc 20 mL로 2회 더** 추출합니다(총 3회). EtOAc 층을 합칩니다(약 60 mL).

☐ 19. 합친 유기층을 깔때기에 넣고 **브라인 15 mL**로 씻습니다. 유기층을 **마른 250 mL 삼각플라스크**에 받습니다.
이유*:* 브라인이 EtOAc에 녹아 있는 물을 끌어내서, 건조제가 처리할 물이 줄어듭니다.

☐ 20. **무수 Na_2SO_4**를 넣고 **15분** 둡니다.

☐ 21. 건조하는 동안 **TLC**: **출발 케톤**과 **반응 혼합물**(건조 중인 용액)을 점적합니다. **헥세인:EtOAc 9:1**로 전개하고 **$KMnO_4$**로 염색합니다. 판 그림에 모든 거리(cm)를 적습니다.
이유*:* 두 화합물 모두 UV로는 잘 안 보이지만 C=C가 있어 $KMnO_4$에 산화됩니다(보라 바탕에 노란 갈색 점). 알코올이 케톤보다 아래에 올 것으로 예상됩니다. 이 판으로 11주차 컬럼 용리액을 정합니다.

☐ 22. **무게를 재 둔 250 mL 둥근바닥 플라스크**에 거르고 **회전증발기**로 용매를 제거합니다 → linalool 조생성물.

☐ 23. **플라스크 + 조생성물** 무게를 재고 사진을 찍은 뒤 datasheet(p. 27)의 **crude product** 칸을 채웁니다.

☐ 24. 11주차까지 보관할 수 있도록 라벨을 붙여 보관합니다. **[CONFIRM: 보관 장소; 마개만 할지 N_2 아래 둘지]**

☐ 25. 사진을 GTA에게 보여 주고 **인쇄한 공식 datasheet**에 확인/서명을 받습니다. **[CONFIRM: 지금 서명인지, 11주차 정제 후 서명인지]**

DATASHEET — WORKING LOG

Synthesis of (±)-Linalool, Part 1 (Grignard reaction, work-up, TLC)

Name: KIM DONGHWAN Lab group: A Fumehood: _____ Date: 15 Oct 2026 Photos taken: □

1. Starting materials

6-Methyl-5-hepten-2-one (mL)	mass = $V \times 0.855$ (g)
n(ketone) = mass/126.20 (mmol)	VinylMgBr solution (mL)
n(vinylMgBr) = $V \times 1.0$ M (mmol)	Equiv. vinylMgBr
Anhydrous THF (mL)	Theoretical (g) = $1.045\, V_{\text{ketone}}$

2. Crude product (datasheet p. 27, "Crude product")

Empty 250 mL rbf (g)	rbf + crude (g)
Crude product (g)	Crude mass / theoretical (%) — *not a yield*

The crude still contains EtOAc and side products; the true yield comes from the purified mass in Week 11.

3. Time log

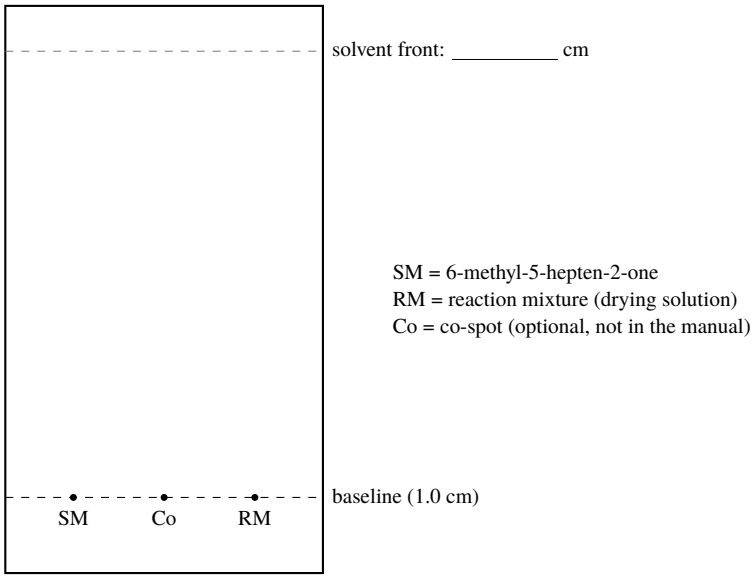
Drying cycle 1 done	Drying cycle 2 done	Ketone + THF added	Grignard addition start	Addition end
Ice bath off (90 min starts)	NH_4Cl quench start	Na_2SO_4 added	TLC run	Rotavap done

4. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Heat-gun drying	no flammables in hood [M]; no visible condensation on glass [E]		
N_2 backfill	bubbler keeps bubbling; no oil sucked back [L]		
Ketone + THF	clear, colourless solution [E]		
VinylMgBr in syringe	*record colour*; only the N_2 buffer, no air bubbles [E]		
During addition	*record any warming, colour change or cloudiness* [E]		
After 90 min at RT	*record colour and any solid (Mg alkoxide)* [E]		
NH_4Cl quench	gas forms (ethylene from excess Grignard) [M]; ≈110 mL [C]; warming; white Mg salts [E]		
Separatory funnel	two layers, EtOAc on top [E]		
Na_2SO_4	some crystals stay free-flowing (solution dry) [E]		
TLC, $KMnO_4$	new spot below the ketone; ketone spot faint/absent if complete [E]		
Crude after rotavap	colourless to pale-yellow oil [E]		

Sources: [M] stated in the lab manual; [L] lecture L4; [C] calculated from manual quantities; [E] general expectation, not stated in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see; a mismatch is a result worth discussing.

5. TLC record (hexane:EtOAc 9:1, $KMnO_4$ stain) — draw the plate and label all distances (cm)



Lane	Spot distance (cm)	Solvent front (cm)	R_f	Colour / intensity
SM				
RM, spot 1				
RM, spot 2				
Co-spot				

6. Carry-over to Week 11 (column chromatography)

Crude stored at / label	Product R_f to target in the column
Column eluent (manual): 200 mL hexane:EtOAc 9:1	Bring: this TLC record + the crude flask

7. Other observations

To be completed at the end of the session		
Student signature	GTA name	GTA signature

This is a working log. The **official** datasheet (next page; lab manual p. 27) must still be printed, signed by the GTA and submitted with the report. It has both a crude and a purified section; the purified section is filled in Week 11.

Name: _____

Lab Group_____ Fumehood _____ Date of the experiment: _____

Datasheet for Synthesis of (±) Linalool using Grignard Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials:

- Amount of 6-methyl-5-hepten-2-one used (mL)_____
- Amount of vinyl magnesium bromide used (mL)_____

Crude product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Purified product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Observations

Draw your TLC plate here and label all distances (in cm)

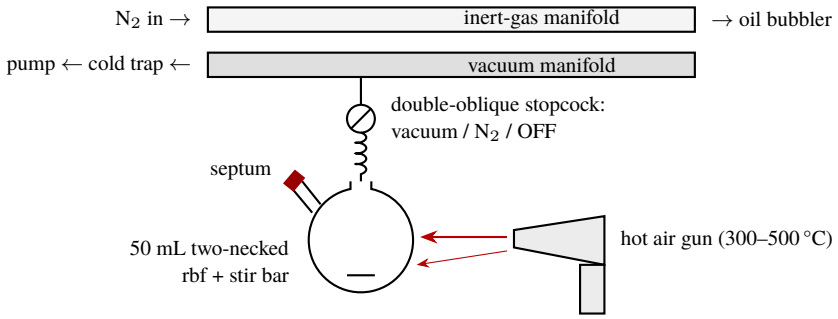
APPENDIX A — APPARATUS

Schlenk-line drying, syringe transfer, dropwise addition and extraction

Schematic, not to scale.

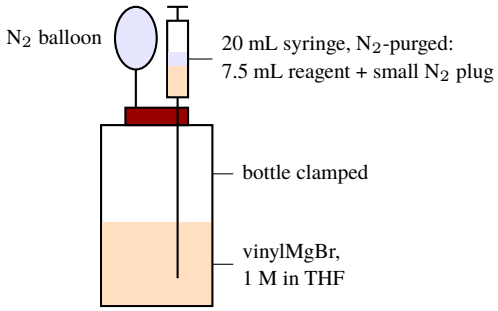
---

(1) Schlenk line: heat-gun drying under vacuum



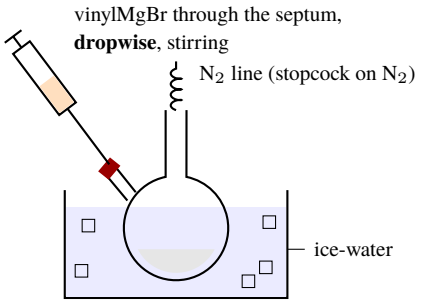
cycle: vacuum + heat → cool → N_2 (**slowly**) ×2 then cool to RT

(2) Withdrawing vinylMgBr under N_2



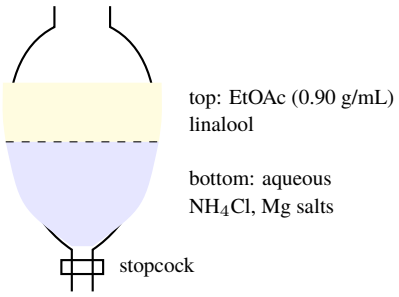
long needle into the liquid

(3) Dropwise addition at 0 °C



then: ice bath off, RT 90 min

(4) Extraction (150 mL separatory funnel)



EtOAc 20 mL × 3 → combine
→ brine 15 mL → Na_2SO_4

Stopcock positions (two-way valve)

Vacuum	flask open to the pump: drying cycles
N_2	flask open to inert gas: backfill (slowly), during the reaction
OFF	flask isolated: before disconnecting for work-up

ORDER OF EVENTS

Hood cleared → vacuum + heat gun → cool → N_2 (**slowly**) ×2 → RT → ketone + THF → ice bath 5–10 min → vinylMgBr 7.5 mL **dropwise** → RT **90 min** (quench syringe, pre-weigh rbf, set up TLC) → valve OFF, disconnect → ice bath, **vent**, sat. NH_4Cl dropwise → water → EtOAc ×3 → brine → Na_2SO_4 15 min + **TLC** → filter → rotavap → **weigh crude** → store for Week 11.

Tip (not in the manual). If the Grignard reagent looks cloudy or has a solid at the bottom of the bottle, tell the GTA before drawing it: a precipitate can mean partial hydrolysis, and less active reagent shows up as unreacted ketone on the TLC.